

SARAL SINGHAL

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Education

Vellore Institute of Technology
Bachelor's in Computer Science (B.Tech)

Bhopal, Madhya Pradesh
2023 – 2027

Maheshwari Public School
(Higher Secondary Education)

Jaipur, Rajasthan
2016 – 2022

Technical Skills

Languages: Java, Python,
Libraries and Frameworks: TensorFlow, Keras, NumPy, Pandas, Matplotlib
Tools: Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Oracle BI

Experience

Ernst & Young
Research Intern

Delhi, India
May 2025 – June 2025

- **Market Analysis:** Performed a detailed assessment of non-fare box revenue streams for Chennai Metro Rail Limited (CMRL), evaluating commercial space utilization and land value capture mechanisms.
- **System Specification:** Designed functional requirements and workflow models for a software platform to automate vendor-based non-fare revenue collection and monitoring.
- **Technical Documentation:** Prepared and presented weekly technical reports and architectural briefs to senior consultants, communicating research insights and proposed system design.

Projects

ChurnSense Python, XGBoost, SMOTE, SHAP

- **Decision Intelligence Framework:** Designed a churn-risk and ROI optimization system quantifying customer lifetime value, retention costs, and campaign success probability to support data-driven business decisions.
- **Strategic Customer Segmentation:** Translated predictive outputs into actionable retention strategies by segmenting customers into value-risk groups, enabling targeted marketing spend and intervention planning.
- **Ensemble Modeling Pipeline:** Built a soft-voting ensemble combining Logistic Regression, Random Forest, and XGBoost with SMOTE balancing and standardized preprocessing for robust churn prediction.
- **Interactive Analytics Platform:** Developed a Streamlit dashboard integrating profit curves, churn-risk scoring, K-Means clustering, and explainability modules for end-to-end churn management.

BleedDetectX Python, TensorFlow, Keras, PyTorch

- **Medical Imaging Models:** Developed deep learning architectures including U-Net, ResNet-UNet, and DMP-WCEBleedNet for bleeding classification and segmentation in Wireless Capsule Endoscopy images.
- **High-Performance Results:** Achieved peak performance using DMP-WCEBleedNet with a Dice Coefficient of 0.812 and classification accuracy of 98.31% on the test dataset.
- **Multi-Task Learning:** Built a PyTorch-based ResNet-UNet leveraging a pre-trained ResNet34 encoder for simultaneous bleeding classification and segmentation.
- **Comprehensive Evaluation:** Deployed DMP-WCEBleedNet with multi-metric assessment including Dice, IoU, Precision, Recall, and F1-Score to ensure clinical-grade reliability.

Certifications

- Microsoft Certified: Data Engineer Associate
- Oracle Analytics Cloud Certified Professional
- Google Skill Badge: Inspect Rich Documents with Gemini Multimodality and Multimodal RAG (Vertex AI)
- Google Skill Badge: Prompt Design in Vertex AI
- Applied Machine Learning Specialization – Coursera